Study of woodwind-like systems through nonlinear differential equations. Part I. Simple geometry

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Idealized woodwind models assume a well-localized nonlinearity coupled to a linear bore whose mathematical description is usually expressed through a convolution product. For the case of the simplest bore geometries, cylindrical and conical, the convolution can be transformed into a delayed differential equation. When including in it the nonlinearity, the usual concepts of nonlinear dynamical systems allow a better understanding of the system's evolution. In this paper they are applied to the cylindrical and conical geometries and some characteristics of their behaviors are analyzed at the threshold of oscillation through an analytical study. It is shown that their dynamics cannot be reduced to a finite number of degrees of freedom (effective dynamics). Finally numerical simulations reveal peculiar characteristics of the direct and inverse bifurcations involved in such simple systems for quadratic and cubic nonlinearities respectively. © *1997 Acoustical Society of America.* [S0001-4966(97)01410-0]

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INTRODUCTION

Traditionally, woodwinds have been represented through idealized models which assume a linear part (the bore) and a point localized nonlinear part (the reed and more generally the excitatory system) at the bore entrance section. Their mutual interaction leading to the self-sustained oscillatory behavior is represented through the diagram shown in Fig. 1. Their time-domain study has been based on algebraic or differential equations for the nonlinear part and on an integral or convolution equation coupling the reed to the linear part:

$$\frac{du(t)}{dt}=F_{\mathrm{NL}}((u(t),p(t),P_0), \tag{1a}$$

$$p(t)=h(t)*u(t), \tag{1b}$$

where F_{NL} is a nonlinear algebraic equation and $h(t)$ is the impulse response at the bore entrance section. This formulation can be very suitable for simulation, but does not give any evident clues to find the values of the parameters leading to all possible behaviors of the system. Particularly, it is not easy to predict under which conditions stable and unstable oscillations will take place and which degrees of freedom govern the amplitude of the oscillation at steady state.

Some integral studies have been carried out (Worman, 1971; Schumacher, 1978; Grand, *et al.*, 1994c), but they often need heavy and complicated calculations partially made in the frequency domain to obtain final results in the time domain. It is not surprising since the frequency domain is not well suited for nonlinear systems.

The study of the stability of a nonlinear dynamical system (NLDS) can be done with the classical tools and concepts of this area provided the system description is made entirely through differential equations. Even if the dynamics of chaos is currently a field of increasing interest, the basic concepts, tools, and experiments date from the nineteenth century. It is not our purpose to give extensive information about this domain, which the reader can easily find in references (Bergé *et al.*, 1984; Lauterborn and Parlitz, 1988). The aim of the following work is to show that for a cylindrical and a conical resonator, Eqs. (1a) and (1b) can be transformed into nonlinear delayed differential equations. Since these geometries are far from real instruments, a second paper is devoted to the case of a real geometry (Barjau and Gibiat, 1997) which is dealt with using a different approach.

In order to transform the convolution equation (1b) into a differential one, it is necessary to have an analytical expression for the kernel $h(t)$. Theoretical results are available for the cylinder (Polack *et al.*, 1987), which accurately fit the experimental data. For the geometries with which we are concerned, an *ad hoc* formulation is also possible which correctly reproduces Polack's results. The nonlinear delayed differential equations derived from this *ad hoc* formulation can then be studied through the concepts of NLDS. The system bifurcations are analyzed from the system simulation with the help of a fast numerical algorithm (Barjau *et al.*, 1990).

I. *AD HOC* FORMULATION FOR THE TIME RESPONSE OF THE CYLINDER AND THE CONE

Various models can be found in the literature for the simplest geometry, the cylinder. If the radiation at the open end and the viscothermal losses are neglected, $h(t)$ reduces to a comb of alternate Dirac impulses (Maganza, 1985). Another model has been used by McIntyre *et al.* (McIntyre *et al.*, 1983) where a Gaussian-type reflection function $r(t)$

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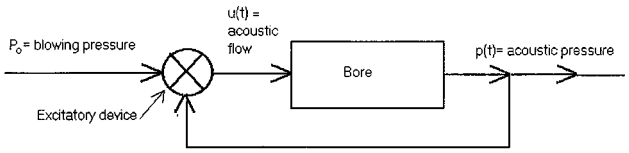

FIG. 1. Schematic diagram of an idealized woodwind. The feedback loop represents the mutual interaction between the linear and the nonlinear parts of the system.

is used instead of $h(t)$. Both models do not respect the basic asymmetry of time-domain responses. More recently Polack *et al.* (Polack *et al.*, 1987) have developed the analytical calculation for the reflection function of a cylindrical pipe. Their result is mathematically complicated but in very good agreement with the experimental data.

The use of the reflection function $r(t)$ instead of $h(t)$ leads to a shorter time-domain description of the cylindrical bore, greatly reducing the time required to compute the convolution equation, which is now written as

$$p(t)-\frac{Z_0}{S}\,u(t)=r(t)*\left[p(t)+\frac{Z_0}{S}\,u(t)\right], \tag{2}$$

and can be seen as a simple change of variables (McIntyre *et al.*, 1983; Maganza, 1985), where Z_0 is the characteristic impedance of the air and S is the bore entrance section area. For other shapes, $r(t)$ is not necessarily shorter but it is always lower than $h(t)$ (Schumacher, 1981).

It is always advisable to work with the minimal response (the shortest possible one) for every possible bore geometry. The reflection function $r(t)$ is the minimal response only for the cylindrical case. For the conical case the minimal response is provided by the use of the so-called modified impulse response $h_{\mathrm{mod}}(t)$ (Agulló *et al.*, 1988). This concept is equivalent to $r(t)$ for the cylindrical case, the only difference being a Dirac $\delta(t)$ distribution and a multiplying constant. The shift from $h(t)$ to $h_{\mathrm{mod}}(t)$ can be seen from a mathematical point of view as a change of variables:

old variables: $(p(t),u(t))$

old equation: $p(t)=h(t)*u(t)$;

new variables: $(p(t),n(t))$

new equation: $p(t)=h_{\mathrm{mod}}(t)*n(t)$.

Both the cone and the cylinder share the same $h_{\mathrm{mod}}(t)$, although the function $n(t)$ to be convolved with it to obtain $p(t)$ is different. For the cylindrical bore,

$$p(t)=h_{\mathrm{mod}}(t)*n(t)=h_{\mathrm{mod}}(t)*\left[u(t)+\frac{S}{Z_0}\,p(t)\right], \tag{3}$$

and for the conical one,

$$p(t)=h_{\mathrm{mod}}(t)*\left[u(t)+\frac{S}{Z_0}\,p(t)+\frac{S}{Z_0}\,\frac{c}{r}\int_0^t p(t')dt'\right], \tag{4}$$

where r is the distance from the entrance section of the truncated cone to the full cone apex.

We proposed a formulation for $h_{\mathrm{mod}}(t)$ (Grand and co-workers, 1994a, b, c) based on an exponential function

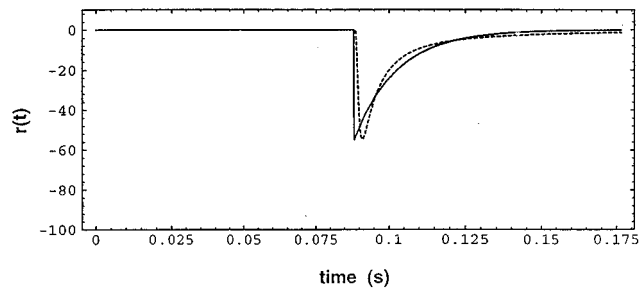

FIG. 2. Comparison between our approach (continuous line) and Polack's formulation (dotted line) for the case of a cylindrical tube of length 15 m and diameter 5 mm, $\sigma=70\ \mathrm{s}^{-1}$, $\beta=55$.

which allows the transformation of the convolution into a differential equation while remaining close to the exact solution given by Polack *et al.* (Polack *et al.*, 1987):

$$h_{\mathrm{mod}}(t)=\frac{1}{2}\,\frac{Z_0}{S}\,[\,\delta(t)-\beta\epsilon(t-T)\exp(-\sigma(t-T))], \tag{5}$$

where $\delta(t)$ and $\epsilon(t)$ are the Dirac and the Heavyside distributions, respectively, T is the time needed for a round trip in the bore, and σ and β are related to the losses and properly adjusted to fit the experimental data.

The choice of an exponential function is not surprising. On the one hand, its mathematical properties (easy derivation and integration) simplify and make possible the analytical manipulations. On the other hand, it is often used to describe the time-extended decay of variables in physical phenomena with damping (as for example the string in a plucked string; Cremer, 1984).

It is clear that this description corresponds neither to the physical frequency-dependent damping due to viscothermal losses, nor to the radiation condition at the open end of the tube usually found in the literature (Bruneau, 1983), but it is a sufficiently accurate description for our purpose and it is an improvement compared to the previous models since it respects the basic asymmetry of the real reflection function. For the case of a cylindrical tube of length 15 m and diameter 5 mm, Fig. 2 shows the comparison between our approach and Polack's formulation with $\beta=55\ \mathrm{s}^{-1}$ and $\sigma=70\ \mathrm{s}^{-1}$. These dimensions have been chosen because they correspond to the example studied theoretically and experimentally by Polack *et al.* They do not correspond to a musical instrument, but the purpose of this article is not the study of a musical instrument—are there musical instruments of absolutely cylindrical or conical shape?—but to work with general systems consisting on a nonlinearity coupled to a duct and to obtain hints on the possible behaviors of musical instruments.

II. NONLINEAR DELAYED DIFFERENTIAL EQUATIONS

The Laplace transform of Eqs. (3) and (4) combined with Eq. (5) yields Eqs. (6) and (7) for the cylinder and the cone, respectively (Grand and co-workers, 1994a, b, c):

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$$\frac{dp(t)}{dt}+\sigma p(t)+\beta p(t-T)$$
$$=\frac{Z_0}{S}\left[\frac{du(t)}{dt}+\sigma u(t)-\beta u(t-T)\right], \tag{6}$$

$$\frac{d^2p(t)}{dt^2}+\left[\sigma-\frac{c}{r}\right]\frac{dp(t)}{dt}-\sigma\frac{c}{r}p(t)$$
$$+\beta\frac{dp(t-T)}{dt}+\beta\frac{c}{r}p(t-T)$$
$$=\frac{Z_0}{S}\left[\frac{d^2u(t)}{dt^2}+\sigma\frac{du(t)}{dt}-\beta\frac{du(t-T)}{dt}\right]. \tag{7}$$

Actually, Eq. (6) can be obtained from Eq. (7) by taking $r\rightarrow\infty$.

These equations are general and apply for any description of the excitatory system as the coefficients β and σ can take any value (according to the chosen tube) and the nonlinear function $u(t)$ has not been specified yet. The only important assumption (or restriction) is that of the one-point spatial localization of the nonlinearity. The main characteristic of these two equations is that they involve delayed values of the variables. In other words, their integration at any time t calls for the knowledge of an infinite set of initial conditions (corresponding to the time evolution of the variables over the previous time interval of length T). So, from a physical point of view, they describe an infinite-degree-of-freedom system which can in principle exhibit any kind of dynamic behavior, from the simple classical ones (static, periodic...) to more complicated ones (quasi-periodic, chaotic,...).

However, from an intuitive point of view, it is clear that if T is small compared to the characteristic times of the physical system (relaxation, damping,...) the number of degrees of freedom can be dramatically reduced. At the limit of $T\rightarrow 0$ Eqs. (6) and (7) would imply just one and two degrees-of-freedom, respectively.

Nonlinear differential equations with delay are not as rare in the nonlinear dynamical systems domain, particularly when studying natural self-oscillating phenomena; to give just a few examples: leukocyte concentration in human blood (Mackey and Glass, 1977), and evolution of animal population in closed ecological systems (May, 1980). They also appear in numerous physical systems: laser (Le Berre *et al.*, 1986), hydrodynamic systems (Villermaux, 1993), etc.

If T is not negligible, it is well-known that the system behavior at steady state is governed by a reduced number of degrees-of-freedom, or modes (Manneville, 1991). According to a classical classification we will talk of ''central modes'' and ''marginal'' or ''slave modes,'' the central modes being those whose damping becomes zero at the threshold of oscillation (thus undergoing a bifurcation) and which govern the amplitude of the marginal ones through a nonlinear relation at steady state (see the Appendix). This reduced description is not always possible, but when it exists, it is known as the system effective dynamics. Of course such a description is meaningless at the transient state, so numerical simulations of steady state are unavoidable to validate the simplification. If the effective dynamical description is not possible, the system oscillations depend on all its modes in principle and then the steady-state amplitude will be limited by the nonlinearity of the system.

We will first consider the simplest possible (quadratic) nonlinear function coupling the flow and the acoustic pressure at the bore entrance section (McIntyre *et al.*, 1983):

$$u(t)=\begin{cases}\alpha[P_0-p(t)][p(t)-p_f], & \text{for } p(t)>p_f,\\ 0, & \text{for } p(t)\leqslant p_f,\end{cases} \tag{8}$$

where P_0 is the blowing pressure, p_f the pressure under which no flow is introduced in the system, and α is a properly adjusted coefficient. A cubic function will be used later on to show the different behavior to which it gives rise.

As we will see in the next section, the lack of description for the reed dynamics does not invalidate the method, as such dynamics can always be introduced through a finite number of additional modes.

III. SYSTEMATIC STUDY OF MARGINAL AND CENTRAL MODES

In order to obtain the effective dynamical description, it is necessary to determine the central or bifurcating modes. They depend in principle on the values of the system parameters. In our study the blowing pressure P_0 (or any function of it) is a natural control parameter since it is one of those that is controlled by the musician and the only one easily measurable.

The systematic study of the effective dynamics begins with the determination of the equilibrium or fixed points, and is followed by a linear analysis at their neighborhood and a search for the bifurcating modes.

A. Cylindrical case

The delayed differential equation (6) together with the definition given for the air flow in Eq. (8) leads to

$$A\left[\frac{dp(t)}{dt}+\sigma p(t)\right]=-\alpha\frac{Z_0}{S}\left[\frac{dp(t)}{dt}+\sigma p(t)\right]p(t)$$
$$+\beta\left[A-2+\alpha\frac{Z_0}{S}p(t-T)\right]$$
$$\times p(t-T)+B, \tag{9}$$

where the parameters A and B are functions of P_0 with $A=1-\alpha(Z_0/S)(P_0+p_f)$ and $B=\alpha(Z_0/S)(\beta-\sigma)P_0p_f$. The equation defining the two equilibrium positions, that is, the values of the internal pressure corresponding to nonoscillatory solutions, $dp/dt=0$, is

$$A(P_0)\sigma p_{\text{eq}}=-\alpha\frac{Z_0}{S}\sigma p_{\text{eq}}^2+\beta\left[A(P_0)-2\right.$$
$$\left.+\alpha\frac{Z_0}{S}p_{\text{eq}}\right]p_{\text{eq}}+B(P_0).$$

The stability of such positions is given by the sign of the time derivative of the pressure:

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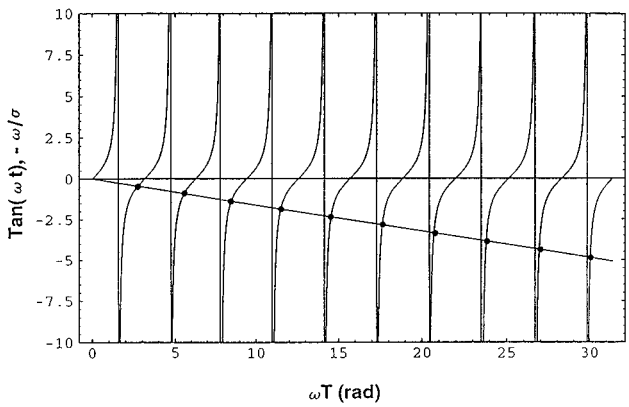

FIG. 3. Graphical solution of the transcendental Eq. (13). The intersection (dots) between $\tan(\omega T)$ and ω/σ (straight line) give the bifurcating modes for the cylinder of Fig. 2.

$$\left.\frac{dp(t)}{dt}\right|_{\mathrm{eq}}=\frac{\beta[\lambda(P_0)-2]-\sigma\lambda(P_0)}{\lambda(P_0)}\,p_{\mathrm{eq}}$$
$$=(\beta-\sigma)-\frac{2\beta}{\lambda(P_0)}. \tag{10}$$

The linear approximation at the neighborhood of p_{eq} [pressure$=p_{\mathrm{eq}}+p(t)$ with $p(t)$ small], leads to

$$\lambda(P_0)\,\frac{dp(t)}{dt}+\sigma\lambda(P_0)p(t)=\beta[\lambda(P_0)-2]p(t-T), \tag{11}$$

with $\lambda(P_0)=A(P_0)+2\alpha(Z_0/S)p_{\mathrm{eq}}$ as the control parameter since it is P_0 dependent.

Substitution of a solution of the form $p(t)=p(0)\exp((\xi+j\omega)t)$ in Eq. (11) splits it into two equations (real and imaginary parts):

$$\lambda(\xi+\sigma)=\beta(\lambda-2)\exp(-\xi T)\cos(\omega T), \tag{12a}$$

$$\lambda\omega=-\beta(\lambda-2)\exp(-\xi T)\sin(\omega T). \tag{12b}$$

The bifurcating modes will be those for which $\xi=0$, and are given by the transcendental equation

$$\tan(\omega T)=-\frac{1}{\sigma T}\,(\omega T). \tag{13}$$

This equation is graphically solved simply by studying the intersection of the two functions defined by its left- and right-hand sides, as presented in Fig. 3 (σ and T correspond to the cylinder of Fig. 2; even if that case is not a musical one it allows a clear visualization).

One important feature of Eq. (13) is that losses appear only on the right-hand side. It is the slope of the straight line which carries this information. For negligible losses ($\sigma\to\infty$) the straight line becomes horizontal and the solutions of Eq. (13) form a complete harmonic set and correspond both to the ''resonances'' *and* to the ''anti-resonances'' of the amplitude impedance curve of the cylinder with an open end. This result points in the same direction as that stated by Grand (Grand, 1994c). As losses grow, the ''resonance'' solutions tend to form a set of odd harmonic values. If losses are infinite ($\sigma=0$), we have just the solution $\omega=0$, which means that no oscillation will take place. Except for the latter case, the number of bifurcating modes is always infinite (although countable) and independent of the control parameter, and the reduction of the model to an effective one is not possible.

This clearly indicates that a perfect cylinder, with a perfectly localized and instantaneous nonlinear part, is, at its threshold of oscillation, a perfectly unstable system where *all* the modes can participate. When σ is infinite, the modes are harmonic so one can expect a periodic solution. When σ is not infinite, the modes are no longer harmonic and the transient will contain growing nonharmonic modes. The periodicity, if any, is then obtained through a phase locking process governed by the nonlinearity at higher amplitude.

Another interesting result which can be obtained straight away from Eqs. (12a) and (12b) is that of the existence of a nonoscillatory mode with damping independent of the control parameter. By setting $\omega\to 0$, we obtain

$$\left.\begin{aligned}&\lambda(\xi_0+\sigma)=\beta(\lambda-2)\exp(-\xi_0 T)\\&\frac{\lambda}{T}=-\beta(\lambda-2)\exp(-\xi_0 T)\end{aligned}\right\}\Rightarrow\xi_0=-\left(\frac{1}{T}+\sigma\right). \tag{14}$$

In order to know what is actually going to happen when we ''blow'' into the cylindrical tube it is necessary to perform numerical simulations: The nonoscillatory mode could lead to a fixed point (equilibrium point) or be ''killed'' by the bifurcating ones. The possibility of a limit cycle cannot be studied analytically as the equilibrium points. Such behaviors can only be determined by numerical simulation. If a limit cycle is found, it is necessarily a stable one, as unstable behaviors are rarely detectable.

B. Conical case

The delayed differential equation (7) together with Eq. (8) yields

$$\frac{d^2p(t)}{dt^2}+\left(\sigma-\frac{c}{r}\right)\frac{dp(t)}{dt}-\sigma\frac{c}{r}\,p(t)$$
$$+\beta\left[\frac{dp(t-T)}{dt}+\frac{c}{r}\,p(t-T)\right]$$
$$=\alpha\frac{Z_0}{S}\left\{[p_f+P_0-2p(t)]\left[\frac{d^2p(t)}{dt^2}+\sigma\frac{dp(t)}{dt}\right]\right.$$
$$\left.-2\left(\frac{dp(t)}{dt}\right)^2-\beta[p_f+P_0-2p(t-T)]\,\frac{dp(t-T)}{dt}\right\}, \tag{15}$$

and setting the derivatives equal to zero, we obtain the equation giving the only possible equilibrium position:

$$(\beta-\sigma)\,\frac{c}{r}\,p_{\mathrm{eq}}=0. \tag{16}$$

The linear approximation at the neighborhood of $p_{\mathrm{eq}}=0$ is

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$$(1-A)\,\frac{d^2p(t)}{dt^2}+\left[\sigma(1-A)-\frac{c}{r}\right]\frac{dp(t)}{dt}-\sigma\,\frac{c}{r}\,p(t)$$
$$=-\beta\left[(1+A)\,\frac{dp(t-T)}{dt}+\frac{c}{r}\,p(t-T)\right], \tag{17}$$

with $A=\alpha(Z_0/S)(p_f+P_0)$ as the only control parameter. The substitution of a solution of the form $p(t)=p(0)e^{(\xi+j\omega)t}$ in Eq. (17) gives two equations (real and imaginary parts):

$$(1-A)(\xi^2-\omega^2+\sigma\xi)-\frac{c}{r}\,(\xi+\sigma)$$
$$=-\beta\,\exp(-\xi T)\left\{\left[\xi(1+A)+\frac{c}{r}\right]\cos(\omega T)+\omega(1+A)\sin(\omega T)\right\}, \tag{18a}$$

$$(1-A)(2\xi\omega+\sigma)-\frac{c}{r}\,\omega$$
$$=-\beta\,\exp(-\xi T)\left\{\omega(1+A)\cos(\omega T)-\left[\xi(1+A)+\frac{c}{r}\right]\sin(\omega T)\right\}, \tag{18b}$$

and we obtain the bifurcating modes by setting $\xi=0$:

$$\tan(\omega T)=-\omega\,\frac{\omega^2(1-A^2)-(c/r)(1-2\sigma)}{\omega^2\sigma(1-A^2)-(c/r)(c/r+2\omega^2)}$$
$$=F(\omega), \tag{19}$$

whose graphical solution is given in Fig. 4(a). For a medium value of the losses, the solutions form a complete family of quasi-harmonic frequencies. As in the case of the cylinder, the reduction to an effective model is impossible.

The nonoscillatory modes are defined by a cubic equation obtained by setting $\omega=0$ in Eqs. (18a) and (18b):

$$(A^2-1)\xi^3+\left[(A^2-1)(1+\sigma)+2\,\frac{c}{r}\right]\xi^2+\frac{c}{r}\left[2(A+\sigma+1)-\frac{c}{r}\right]\xi+\frac{c}{r}\left[\frac{c}{r}\,(1-\sigma)-2\sigma\right]=0. \tag{20}$$

Solution of Eq. (20) yields three roots whose dependence on the control parameter P_0 is shown in Fig. 4(b). Actually, two of them remain negative for the calculated range of blowing pressures (and so correspond to stable modes), while the third one soon becomes positive, giving rise to an unstable mode which can help the bifurcating ones in establishing an oscillation. As in the cylindrical case, the zero amplitude threshold oscillation will be necessarily multimodal with nonharmonic modes, and a periodic steady state will only be possible if the nonlinearity succeeds in the phase locking of those modes.

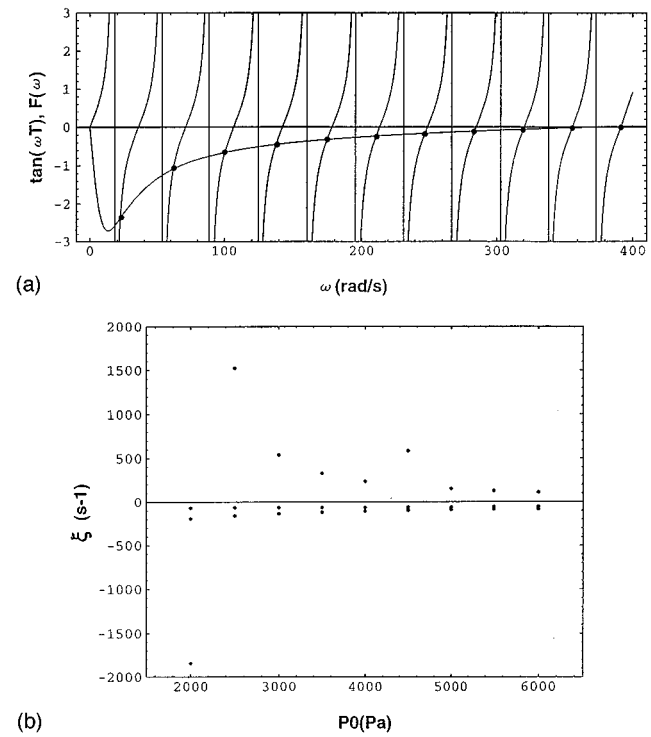

FIG. 4. (a) Graphical solution of the transcendental Eq. (19) giving the bifurcating modes for a cone of length 15 m, with $\sigma=70\ \mathrm{s}^{-1}$. Entrance diameter is 4 mm and final diameter 64 mm. (b) Damping of the nonoscillating modes for the same cone.

IV. TIME-DOMAIN SIMULATION OF CYLINDRICAL AND TRUNCATED CONICAL "WOODWINDS"

As stated in Sec. III A, the actual result of "blowing" into the instrument can only be studied through the numerical integration of Eqs. (9) and (15). As both geometries, cylindrical and conical, are assumed to be described by a time response of the exponential type in this paper, such an integration can be reduced to an iterative algorithm of the form (Barjau *et al.*, 1990):

$$p(t+\Delta t)=a_1p(t)+a_2n(t)+b_1p(t-\Delta t)+b_2n(t-\Delta t)+c_1p(t-T)+c_2n(t-T)+d_1p(t-\Delta t-T)+d_2n(t-\Delta t-T), \tag{21}$$

where $a_{1,2}$, $b_{1,2}$, $c_{1,2}$, $d_{1,2}$ are constant coefficients and $n(t)$ is the function defined in Sec. I.

The physical information that can be obtained through simulation is not only the value P_0 at the threshold of oscillation (that is, the value at which a first bifurcation appears), but also the type of bifurcation that the system undergoes—direct or inverse.

It has been shown that this type of bifurcation is related to the particular shape of the nonlinear function chosen to represent the excitatory system. A quadratic function implies an inverse bifurcation—discontinuous jump from zero to finite amplitude oscillation—while a cubic one produces a direct bifurcation—continuous transition from zero to finite amplitude of oscillation. This has been noted first by McIntyre *et al.* (McIntyre *et al.*, 1983), although described with different words, explained by Maganza *et al.* (Maganza

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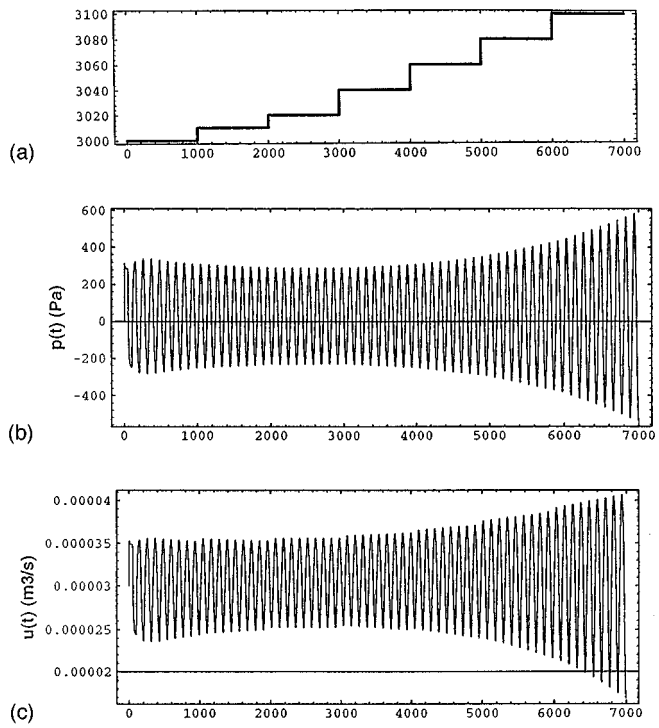

FIG. 5. Pressure (b) and flow (c) obtained with P_0 values (a) increasing from 3000 Pa to 3100 Pa and with a quadratic nonlinearity for the cylinder of Fig. 2.

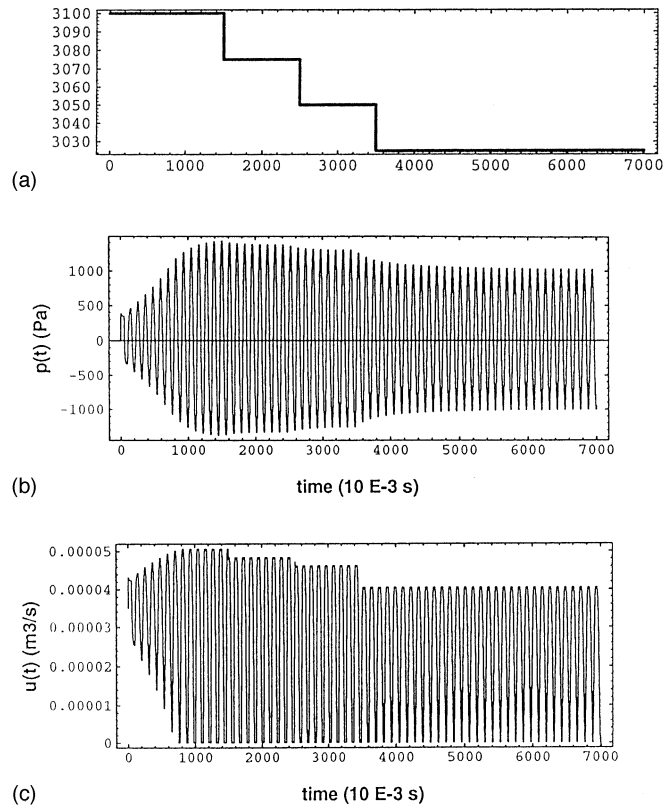

FIG. 6. Pressure (b) and flow (c) obtained with P_0 values (a) decreasing from 3100 Pa to 3000 Pa and with a quadratic nonlinearity for the cylinder of Fig. 2.

et al., 1987) and proved by Grand (Grand *et al.*, 1994c). If the variation of the control parameter leads to a direct bifurcation, a stable oscillation with very low infinitesimal amplitude appears after the threshold. If the bifurcation taking place is an inverse one, the resulting oscillation after threshold has a finite amplitude. Another interesting difference is the existence of an hysteretic phenomenon when the control parameter is decreased for an inverse bifurcation: The oscillation can be maintained with a value of the control parameter lower than the threshold one.

Since a musical instrument is supposed to be played between pianissimo and fortissimo levels, some former works have examined for a direct bifurcation in woodwinds. For the case of the cylinder defined in Fig. 2 and a quadratic nonlinear function, Fig. 5(b) and (c) shows the pressure $p(t)$ and air flow $u(t)$ at the entrance section, respectively, the results for an interval of P_0 values going from 3000 Pa to 3100 Pa [Fig. 5(a)]. The threshold of oscillation is located at about 3050 Pa.

Figure 6(b) and (c) shows $p(t)$ and $u(t)$ at the same section but decreasing values of P_0 going from 3100 Pa to 3000 Pa [Fig. 6(a)]. The graphics clearly show a hysteretical behavior of the system, as a stable oscillation can be attained at $P_0 = 3025$ Pa. That is, the bifurcation leading to a limit cycle from an equilibrium position is an inverse one and the width of the P_0 zone where the hysteretical behavior takes place, ΔP_0, is about 30 Pa (1% of P_0). Figure 7(a) and (b) shows the corresponding bifurcation diagram. The low value of ΔP_0 [Fig. 7(b)] indicates however that the bifurcation is not far from being a direct one.

If the nonlinearity is a cubic one,

$$u(t) = \alpha(P_0 - p(t))(p(t) - p_f)(P_0 + p(t) - 2p_f),$$

the bifurcation becomes a direct one. Simulations for increasing and decreasing values of P_0 have been carried out and do not reveal any hysteresis. Figure 8(a) and (b) presents

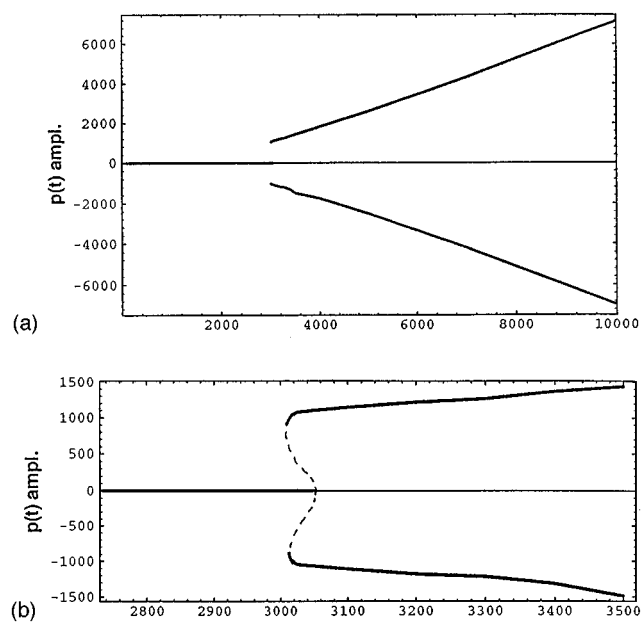

FIG. 7. Inverse bifurcation diagram for a range of P_0 values going from 0 Pa to 10 000 Pa (a) and from 2750 Pa to 3500 Pa (b) for the cylinder of Fig. 2 and the quadratic nonlinearity (the dotted line represents the unstable branches).

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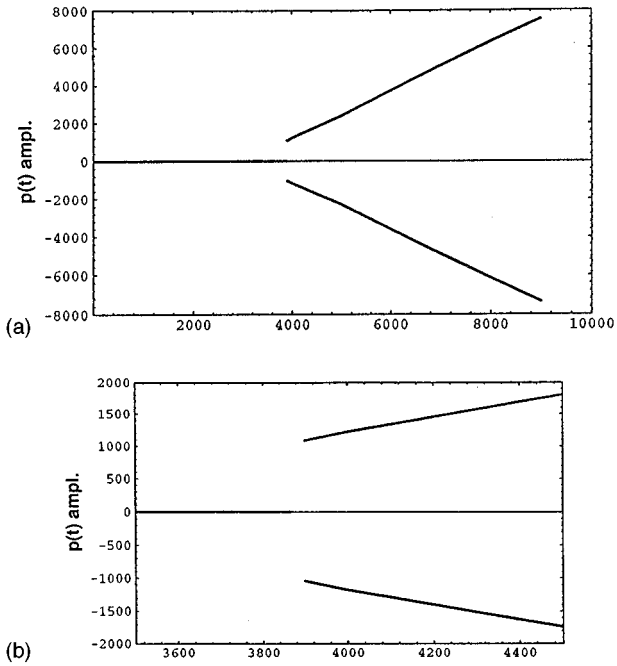

FIG. 8. Direct bifurcation diagram for a range of P_0 values going from 0 Pa to 10 000 Pa (a) and from 3500 Pa to 4500 Pa (b) for the cylinder of Fig. 2 and the cubic nonlinearity.

the bifurcation diagram in this case. Nevertheless, comparison of Figs. 7(a) and 8(a) makes it evident that the difference, even if interesting and certainly from a physical point of view, is perfectly negligible from a ''musical'' point of view.

For the case of the truncated cone, the stability of the simulation is much more dependent on the choice of the parameters. For the case of the quadratic nonlinearity, the threshold of oscillation has been found to be located in the neighborhood of $P_0 = 5200$ Pa. Figure 9 shows the first oscillations for the smooth conicity case of Fig. 4 and a blowing pressure of 5300 Pa. Other simulations have been performed for a wide range of blowing pressures, and their results are schematized in the bifurcation diagram in Fig. 10. It is a direct bifurcation.

Another collection of simulations has been calculated for a different value of the α parameter in the flow function. It has been observed that, after periodic behaviors, the signal becomes biperiodic (second bifurcation), as it can be easily seen in Fig. 11(a) (beating frequency) or in the two other

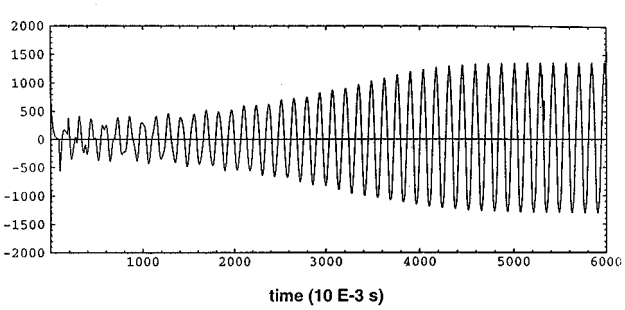

FIG. 9. Pressure obtained with a P_0 of 5400 Pa for the cone of Fig. 4 and the quadratic nonlinearity.

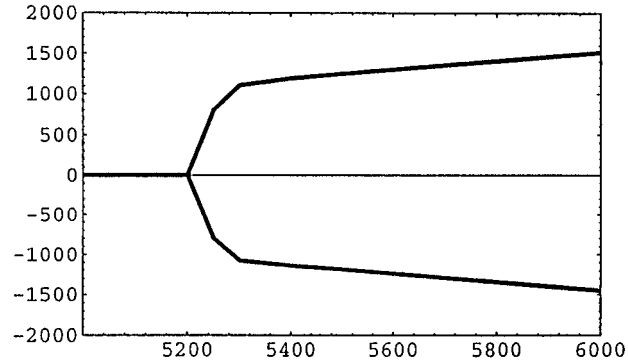

FIG. 10. Bifurcation diagram as a function of the blowing pressure for the cone of Fig. 4 and the quadratic nonlinearity.

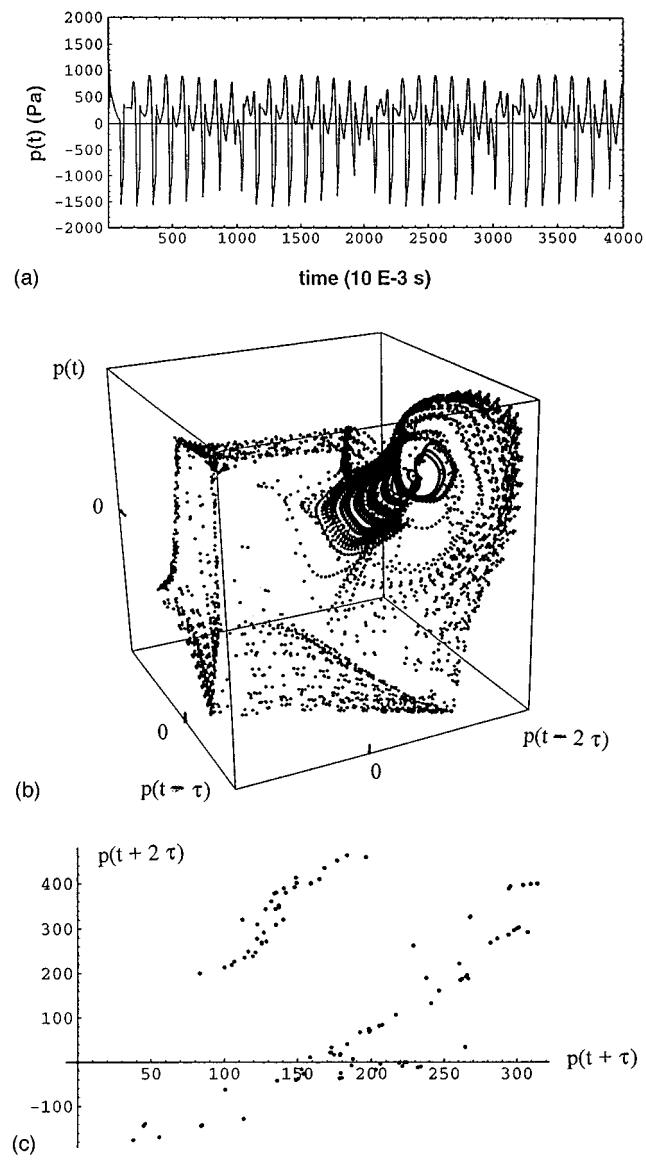

FIG. 11. Simulation of the cone in Fig. 4 for $P_0 = 5000$ Pa and $\alpha = 10^{-10}$. (a) Time amplitude representation. (b) Phase space reconstruction with axes $p(t)$, $p(t-\tau)$, $p(t-2\tau)$, with τ equal to a quarter of the fundamental period of the oscillation. (c) Poincaré section by the plane $p(t)=0$ of the phase space reconstruction.

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representations, phase space reconstruction [toruslike shape, Fig. 11(b)] and Poincaré section [Fig. 11(c)]. This result is not surprising since the delayed differential equation for the cone contains three different characteristic times. The first one is the delay associated with the round trip of the acoustical wave. The second one is the characteristic time of the damping, very small compared to the former and which does not seem to be relevant in the biperiodic state. The third one is the inverse of the eigenfrequency appearing in the nondelayed part of the left-hand side of Eq. (17):

$$\frac{\sigma c}{r(1-A)}.$$

Changes in the conicity (changes in r) affect this characteristic time and thus the signal, although still biperiodic, will show a different beating frequency.

As conicity increases, the simulation becomes more and more sensitive to small changes in the parameters, and the accumulation of numerical truncation errors results in diverging nonphysical behaviors which can only be avoided by dramatically reducing the sampling interval, increasing the numerical precision, and thus highly increasing the required computation time.

V. CONCLUSION

An analytical formulation of the time responses of strictly cylindrical and conical woodwinds has been presented which leads to a totally differential formulation, whose main characteristic is that it includes delayed variables. Such a description is quite new in the domain of musical acoustics even if frequent in other scientific domains.

The study of stability of the system through a linear approach leads to transcendental equations which show the impossibility of reducing the description of such systems to an effective dynamical one. However, they give important information concerning the modes which play an important role at the threshold of oscillation. For the cylindrical case, the even modes participate with a strength comparable to that of odd ones. This result is obtained in a simpler way in the presented time approach than in previous frequency-domain methods (Grand, 1994c). For the conical tube, all modes (even and odd) participate in the oscillating regime. This means that the threshold of oscillation of perfectly conical or cylindrical systems involves all of their resonant modes. Consequently, since truncated conical bores present inharmonic modes, their transient state at any blowing pressure will contain nonharmonic components, as is easily seen in Fig. 9.

A numerical integration of a cylindrical and a conical system has been carried out for different values of the blowing pressure and for two different nonlinear functions (representing the excitatory system coupled to the bore): quadratic and cubic ones. Whenever the bore is a cylindrical one, the results show that a periodic oscillatory regime is attained through an inverse bifurcation in the quadratic case, while the bifurcation becomes direct if the quadratic nonlinearity is replaced by the cubic one. However, the difference between the direct and the inverse bifurcations is so small that it is, in practice, hardly noticeable.

For the case of the studied conical bore, the bifurcation is direct for the quadratic flow. In the parameter space of the system, different control parameters other than the blowing pressure can be used. If the α parameter appearing in the flow function is decreased, a second bifurcation leading to quasi-periodic behavior appears.

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APPENDIX

For a general unidimensional physical system described through a nonlinear partial differential equation of the type:

$$a\,\frac{\partial^i \Phi(x,t)}{\partial t^i}+b\,\frac{\partial^j \Phi(x,t)}{\partial x^j}=F_{\mathrm{NL}}\left(\frac{\partial^k \Phi(x,t)}{\partial t^k},\,\frac{\partial^l \Phi(x,t)}{\partial x^l}\right), \tag{A1}$$

and given initial and boundary conditions, a modal study at the threshold can be done by linearizing the differential equation and trying a separable solution of the form:

$$\Phi(x,t)=T(t)X(x). \tag{A2}$$

The boundary conditions lead to restrictions on $X(x)$, so that $\Phi(x,t)$ becomes a superposition of an infinite number of countable solutions:

$$\Phi(x,t)=\sum_{n=1}^{\infty} T_n(t)X_n(x) \tag{A3}$$

[with well-defined $X_n(x)$]. The time-varying amplitudes $T_n(t)$ are governed by an infinite number of ordinary nonlinear differential equations which can always be expressed, through a suitable change of variables, as first-order ones:

$$\frac{dA_n(t)}{dt}+\sigma_n A_n(t)=F_{\mathrm{NL}}(A_1(t),A_2(t),\ldots,A_m(t)). \tag{A4}$$

A solution to these equations is given by Manneville (1990):

$$A_n(t)=A_n^0 e^{-\sigma_n t}+e^{-\sigma_n t}*F_{\mathrm{NL}}(A_1(t),A_2(t),\ldots,A_m(t)). \tag{A5}$$

It is clear that for $\sigma_n>0$, both terms become negligible for increasing values of t, that is, far from the transient state there are no changes in $A_n(t)$, and Eq. (A4) becomes an algebraic one:

$$A_n(t)=\frac{1}{\sigma_n}F_{\mathrm{NL}}((A_1(t),A_2(t),\ldots,A_m(t))). \tag{A6}$$

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